

5 (a) (i) Current is the rate of flow of charge

(ii) $I = \frac{Ne}{t}$

$$I = \frac{nALe}{t}$$

$$I = nAve$$

(iii) 1. $R = \frac{\rho l}{A}$
 $= \frac{(1.68 \times 10^{-8})(3)}{\pi(0.5 \times 10^{-3})^2}$

$$= 0.064 \Omega$$

2. $V = IR$

$$I = \frac{0.2}{0.064}$$
$$= 3.13 A$$

$$\begin{aligned}
 I &= nAve \\
 v &= \frac{I}{nAe} \\
 &= \frac{3.13}{(8.49 \times 10^{28})\pi(0.5 \times 10^{-3})^2(1.6 \times 10^{-19})} \\
 &= 2.93 \times 10^{-4} \text{ ms}^{-1}
 \end{aligned}$$

(iv) Current is a horizontal straight line

Drift velocity higher than B than A

Increasing gradient

(b) (i)
$$\begin{aligned}
 R_t &= \frac{30}{3.6}(0.4) \\
 &= 3.33\Omega
 \end{aligned}$$

Read off from graph = 59-61 °C

(ii) Effective resistance across the alarm and flashing light will decrease

The resistance of the thermistor will therefore have to be even lower to trigger the alarm, therefore an even higher temperature will be required

(iii) Smoke detector will alert the user before the fire starts